BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: DIK-SSC-IX-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 15 | **Paper Code:** DIK-SSC-IX-2021 (A)

Q1. Choose the correct option:

Q1. The matrix [1 0; 0 3] is called _____

(a) Scalar
(b) Diagonal
(c) Identify
(d) Rectangular

Q2. $\sqrt[3]{-64}$ = _____

(A) −8
(B) −4
(C) 4
(D) 8

Q3. 0.00729 in scientific notation can be written as _____

(a) 729×10^{-3}
(b) 72.9×10^{-5}
(c) 7.29×10^{-3}
(d) 0.729×10^{-3}

Q4. (7x + 4y) − (3x − 6y) = _____

(A) (4x+10y)
(B) (4x−2y)
(C) (4x+2y)
(D) (4x−3y)

Q5. Factorization of $x^2 - 5x + 6$ is_____

(a) (x − 2)(x + 3)
(b) (x − 2)(x − 3)
(c) (x + 2)(x + 3)
(d) (x + 2)(x − 3)

Q6. HCF of a^3-b^3 and $a^2 + ab + b^2$ is _____

(a) a + b
(b) a − b
(c) (a − b)(a + b)
(d) $a^2 + ab + b^2$

Q7. $\sqrt{(2x+1)} - 5 = 4$ is a _____ equation.

(a) Linear
(b) Cubic
(c) Radical
(d) Quadratic

Q8. The two coordinate axes intersect at an angle of _____

(A) 30^0
(B) 45^0
(C) 60^0
(D) 90^0

Q9. All are the sufficient condition for the congruency of two triangles EXCEPT _____

(a) A.S.A ≅ A.S.A
(b) H.S ≅ H.S
(c) S.A.A ≅ S.A.A
(d) A.A.A ≅ A.A.A

Q10. If sum of the measure <A and <C of a parallelogram ABCD is 120, then m<B=_____

(A) 25^0
(B) 50^0
(C) 60^0
(D) 65^0

Q11. The angle bisectors of a triangle meet at a point which is equidistant from _____ of the triangle.

(a) Vertices
(b) Sides
(c) Mid-point of sides
(d) All of these

Q12. A segment bisector is a line that divides a segment into _____ congruent parts.

(A) 4
(B) 3
(C) 2
(D) 1

Q13. In $\triangle ABC$, $m\angle A=50^0$ and $m\angle B=30^0$, which of the following is correct?

(a) $m\bar{BC} > m\bar{AB}$
(b) $M\bar{AB} > m\bar{CA}$
(c) $M\bar{BC} < m\bar{CA}$
(d) $m\bar{AD} < m\bar{CA}$

Q14. Which of the following are the sides of a right angled triangle?

(A) 2,3,4
(B) 3,4,5
(C) 4,5,6
(D) 5,6,7

Q15. Perpendicular from a vertex of a triangle to its opposite side is called _____

(a) Median
(b) Perpendicular bisector
(c) Altitude
(d) Angle bisector

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36 | **Paper Code:** DIK-SSC-IX-2021 (A)

Q2. Attempt any 9 questions of the following:

(i) Solve the system of linear equations $2x + 3y = -1$, $x - y = 2$ with the help of matrics.

(ii) Simplify: $\sqrt{(625x^3y^4/25xy^2)}$

(iii) Simplify $2.83/(6.52)^2$ with the help of logarithm.

(iv) Find the value of $ab + bc + ca$, when $a + b + c = 14$ and $a^2 + b^2 + c^2 = 78$

(v) Factorize $24x^3 + 3$

(vi) Find the square root of $x^2 - 2x + 1 + 2xy - 2y + y^2$

(vii) Find the solution set of $\sqrt{(2x-7)} + 8 = 11$

(viii) Draw the graph of the equation $y - 2x = 6$

(ix) Prove that A(−4,−3), B(1,4) and C(6,11) are collinear.

(x) The sum of three consecutive odd integers is 81. Find the numbers.

(xi) A line parallel to one side of a triangle, intersecting the other two sides divides them proportionally,

(xii) A ladder whose foot is 2.5m from the front of a house reaches a window 6m above the ground. Calculate the length of a ladder.

SECTION – C

Maximum Marks: 24

Q3. Prove that points A(−2,−2), B(4,−2),C(4,6) are the vertices of a right angled triangle.

Q4. If two angles of a triangle are congruent, then the side opposite to these angles are congruent?

Q5. Any point on the right bisectors of a line segment is equidistant from end points of the segment.

Q6. Construct $\triangle xyz$, when $my\bar{z} = 4.1cm$, $m\angle y = 60^o$ and $m\angle x = 75^o$